The Awareness-Cognition Kernel: Ten Tier-2 Theorems on the Structure of Mind Across Phylogeny

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We construct a 10-channel trace vector that partitions cognition into two functional subspaces — *awareness* (reflective, representational capacities) and *aptitude* (somatic fitness) — and apply the domain-agnostic Generative-Collapse Dynamics (GCD) kernel $K\colon [0,1]^{10} \times \Delta^{10} \to (F, \omega, S, C, \kappa, \mathrm{IC})$ to 34 organisms spanning bacteria to human developmental stages. Ten theorems are proven computationally (55/55 subtests, 0 failures): awareness and aptitude are anti-correlated across phylogeny at $\rho = -0.71$ (T-AW-1); no organism with measurable awareness disparity reaches the Stable regime (T-AW-2); aptitude channels are the geometric-mean bottleneck for 84.6% of high-awareness organisms (T-AW-3); the sensitivity formula $\mathrm{dIC}/\mathrm{d}c_k = \mathrm{IC} \cdot w_k/c_k$ is verified to exact precision with an 8.2× ratio for the human adult (T-AW-4); the awareness–aptitude suppression pattern is structurally isomorphic to Standard Model confinement (T-AW-5); the fractional integrity cost Δ/F declines monotonically with awareness (T-AW-6); human development follows a non-monotonic arc peaking at adulthood with $F = 0.637$ (T-AW-7); the binding gate transitions from ω to C at high awareness (T-AW-8); the kernel's F-ranking is preserved across independent closure domains (T-AW-9); and exact analytic bounds $F = (\overline{A}_w + \overline{A}_p)/2$, $\mathrm{IC} = \sqrt{\overline{A}_w \cdot \overline{A}_p}$ are verified to residuals $< 3.3 \times 10^{-16}$ (T-AW-10). All results derive from Axiom-0 through frozen Tier-1 identities ($F + \omega = 1$, $\mathrm{IC} \le F$, $\mathrm{IC} = e^{\kappa}$) applied to biologically motivated channel assignments. The formalism provides a *measurement substrate* for cognition: what can be said about awareness is exactly what survives the kernel.

I. INTRODUCTION

What is awareness? The question has generated millennia of philosophical dispute, largely because the concept resists operationalization: any definition risks either circularity (defining awareness in terms of its products) or category error (reducing subjective experience to a physical observable). William James called consciousness "the one thing that psychology should never eliminate" [1]; Thomas Nagel argued that no objective account can capture "what it is like" to be a conscious subject [2].

We take a different approach. Rather than defining awareness and then seeking its correlates, we define a *measurement substrate* — a domain-agnostic kernel function whose inputs are measurable quantities and whose outputs are structural invariants — and ask what happens when the inputs encode cognition. The question is not what awareness *is*, but what a measurement formalism can *say* about it.

The substrate is the Generative-Collapse Dynamics (GCD) kernel [3, 4], built on a single axiom: *"Collapse is generative; only what returns is real"* [5]. The kernel $K : [0,1]^n \times \Delta^n \to (F, \omega, S, C, \kappa, \mathrm{IC})$ maps any set of coherence coordinates to six invariants satisfying three provably exact identities:

$$F + \omega = 1, \quad \mathrm{IC} \le F, \quad \mathrm{IC} = e^{\kappa}. \tag{1}$$

These identities hold by construction for any input. The kernel does not know what its inputs represent. It computes the same invariants, the same bounds, the same regime classification whether the trace vector encodes particle quantum numbers [6], nuclear binding energies [7, 8], financial portfolios, or — as we demonstrate here — the cognitive capacities of living organisms.

Our contribution is a Tier-2 closure [3] that maps cognition to a 10-channel trace vector with a 5+5 partition: five channels for *awareness* (reflective, representational capacities) and five for *aptitude* (somatic fitness: what the body can do). We apply this closure to 34 organisms spanning four billion years of evolution — from *E. coli* to human developmental stages — and prove ten theorems about the resulting kernel structure.

The central finding is an *inversion*: across phylogeny, awareness and aptitude are strongly anti-correlated ($\rho = -0.71$, $p = 2.4 \times 10^{-6}$). Organisms that invest in reflective capacity systematically sacrifice somatic fitness. The kernel makes this tradeoff *measurable*: it appears as the heterogeneity gap $\Delta = F - \mathrm{IC}$, which quantifies how much fidelity is lost to channel heterogeneity. The gap is driven by the 5+5 partition structure — awareness channels high, aptitude channels low (or vice versa) — and its consequences propagate through every kernel invariant.

This is not a theory of consciousness. It is a measurement framework that encodes what can be measured about cognition into a structure where the consequences are computable, testable, and comparable across species and across domains. The kernel provides the diagnostic lens; the ten theorems describe what is visible through it.

Reader's guide. This paper is self-contained for readers outside the GCD/UMCP framework. The essential machinery is a *kernel* function K (Definition 1) that takes

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any set of normalized scores $c_i \in [0,1]$ with weights w_i summing to 1 and returns six numbers: a weighted arithmetic mean F (*fidelity* — how much signal survives), its complement $\omega = 1 - F$ (*drift* — how much is lost), a weighted geometric mean IC (*integrity* — multiplicative coherence), an entropy S, a curvature C, and the logarithm $\kappa = \ln \mathrm{IC}$. The key diagnostic is the *heterogeneity gap* $\Delta = F - \mathrm{IC} \geq 0$: by the inequality of arithmetic and geometric means it is always non-negative, and it vanishes only when all channels are equal. *Regime classification* bins each organism into Stable, Watch, or Collapse by comparing (F, ω, S, C) against frozen thresholds. No other jargon is needed to follow the ten theorems.

II. KERNEL REVIEW

Definition 1 (GCD Kernel)**.** *Given coordinates* $\mathbf{c} = (c_1, \ldots, c_n) \in [\varepsilon, 1-\varepsilon]^n$ *and weights* $\mathbf{w} = (w_1, \ldots, w_n)$ *with* $\sum_i w_i = 1$*, the kernel computes:*

$$F = \sum_i w_i\, c_i, \tag{2}$$

$$\omega = 1 - F, \tag{3}$$

$$S = -\sum_i w_i \big[c_i \ln c_i + (1-c_i)\ln(1-c_i)\big], \tag{4}$$

$$C = \frac{1}{0.5}\,\mathrm{std}(\mathbf{c}), \tag{5}$$

$$\kappa = \sum_i w_i \ln(c_i + \varepsilon), \tag{6}$$

$$\mathrm{IC} = \exp(\kappa). \tag{7}$$

F is the *fidelity* — the weighted arithmetic mean of channels: how much survives collapse. ω is the *drift* — how much is lost. IC is the *integrity coefficient* — the weighted geometric mean: multiplicative coherence. The heterogeneity gap $\Delta \equiv F - \mathrm{IC} \geq 0$ is the central diagnostic. It vanishes if and only if all channels are equal; it grows with channel disparity.

Regime classification uses four conjunctive gates on the frozen thresholds [3]:

$$\begin{aligned}
&\text{Stable: } \omega < 0.038 \wedge F > 0.90 \wedge S < 0.15 \wedge C < 0.14,\\
&\text{Watch: } 0.038 \leq \omega < 0.30,\\
&\text{Collapse: } \omega \geq 0.30,\\
&\text{Critical: } \mathrm{IC} < 0.30 \text{ (overlay)}.
\end{aligned} \tag{8}$$

The *binding gate* is the gate with the smallest margin to its threshold — the single constraint that most tightly binds the organism away from stability.

III. TRACE-VECTOR CONSTRUCTION

Each organism is mapped to a 10-channel trace vector $\mathbf{c} \in [\varepsilon, 1-\varepsilon]^{10}$ with equal weights $w_i = 1/10$. The channels are partitioned into two functional subspaces of five channels each.

A. Awareness Subspace (Channels 0–4)

The awareness subspace encodes reflective, representational capacities — what the organism *knows* about itself and its world:

TABLE I. Awareness channels (0–4). Each capacity is normalized to $[\varepsilon, 1-\varepsilon]$ using cross-species behavioral assays.

i	Channel	Anchoring
0	mirror_recognition	Self-recognition in the mirror test [9]; 0.01 = chemotaxis, 0.95 = human adult
1	metacognitive_accuracy	Confidence calibration and uncertainty monitoring [10]; 0.01 = fixed response, 0.90 = human
2	planning_horizon	Temporal depth of future-directed behavior [11]; 0.01 = reactive, 0.92 = multi-year
3	symbolic_depth	Recursive symbolic/linguistic capacity [12]; 0.01 = none, 0.98 = recursive grammar
4	social_cognition	Theory of mind, empathic coordination [13]; 0.01 = no ToM, 0.95 = human

B. Aptitude Subspace (Channels 5–9)

The aptitude subspace encodes somatic fitness — what the body can *do*, independent of whether the organism is aware of it:

C. The 5+5 Partition

The partition is not decorative. Define the subgroup means

$$\overline{A}_w = \frac{1}{5}\sum_{i=0}^{4} c_i, \qquad \overline{A}_p = \frac{1}{5}\sum_{i=5}^{9} c_i. \tag{9}$$

When all channels within each subgroup equal their subgroup mean (the *uniform-subgroup approximation*),

TABLE II. Aptitude channels (5–9). Values are benchmarked against species-optimal performance within each channel.

i	Channel	Anchoring
5	sensory_acuity	Peak sensory discrimination (species-optimal); 0.15 = planarian, 0.95 = mantis shrimp
6	motor_precision	Fine motor control relative to body plan; 0.10 = planarian, 0.85 = octopus
7	environmental_tolerance	Range of survivable conditions; 0.15 = human elderly, 0.85 = *E. coli*
8	reproductive_output	Fecundity × viability; 0.10 = human elderly, 0.98 = *E. coli*
9	somatic_resilience	Recovery, longevity, stress response; 0.20 = human elderly, 0.98 = planarian

the kernel outputs reduce to closed-form expressions:

$$F = \frac{\overline{A}_w + \overline{A}_p}{2}, \tag{10}$$

$$\mathrm{IC} = \sqrt{\overline{A}_w \cdot \overline{A}_p}, \tag{11}$$

$$\Delta = \frac{\overline{A}_w + \overline{A}_p}{2} - \sqrt{\overline{A}_w \cdot \overline{A}_p}, \tag{12}$$

$$\frac{\mathrm{IC}}{F} = \frac{2\sqrt{\overline{A}_w \cdot \overline{A}_p}}{\overline{A}_w + \overline{A}_p}. \tag{13}$$

These are identities of the kernel function, not approximations. They hold because F is a weighted arithmetic mean (which, with equal weights and a balanced partition, equals the arithmetic mean of the subgroup means) and $\mathrm{IC} = \exp(\sum w_i \ln c_i)$ is a weighted geometric mean (which factors into the geometric mean of uniform subgroup values). Theorem T-AW-10 verifies these to residuals $< 3.3 \times 10^{-16}$.

The *coupling efficiency* IC/F (Eq. 13) has a well-known form: it is maximal (= 1) when $\overline{A}_w = \overline{A}_p$ and decreases monotonically as the subgroup means diverge. This is the kernel's diagnostic for the awareness–aptitude tradeoff: the more an organism specializes in one subspace at the expense of the other, the lower its coupling efficiency.

IV. ENTITY CATALOG

We catalog 34 organisms across 12 clades, from *E. coli* (Bacteria) through Protist, Nematoda, Insecta, Arachnida, Cephalopoda, Crustacea, Actinopterygii, Reptilia, Aves, Mammalia, and Primates, to nine *Homo sapiens* states (infant, child 5, adolescent, adult, meditator, savant, elderly 65, elderly 85, minimally conscious). Channel values are sourced from published comparative cognition and behavioral ecology literature; see Table III for the full listing.

Key structural features visible in Table III:

- No entity reaches the Stable regime. The minimum drift is $\omega = 0.363$ (human adult), still 9.6× above the threshold $\omega < 0.038$.
- 15 entities are Critical ($\mathrm{IC} < 0.30$); 19 are Collapse ($\omega \geq 0.30$).
- The binding gate is ω for 28 entities, C for 5 (all high-awareness humans), and S for 1 (human adolescent).
- The fractional integrity cost Δ/F ranges from 4.5% (African grey) to 77.9% (*E. coli*).

V. TEN THEOREMS

Each theorem is a computationally verified statement about the kernel outputs for the 34-entity catalog. All are proven with 55/55 subtests and 0 failures. The reference implementation is publicly available [14].

A. T-AW-1: Awareness-Aptitude Inversion

Theorem 2 (Anti-Correlation)**.** *Across the 34-entity catalog, the Spearman rank correlation between awareness mean $\overline{A}_w$ and aptitude mean $\overline{A}_p$ satisfies*

$$\rho(\overline{A}_w, \overline{A}_p) = -0.712, \quad p = 2.4 \times 10^{-6}. \tag{14}$$

Furthermore, awareness dominates fidelity: $\rho(\overline{A}_w, F) = +0.945$ ($p \approx 4 \times 10^{-17}$), while aptitude is negatively correlated: $\rho(\overline{A}_p, F) = -0.525$ ($p = 0.0014$).

Remark 1. *The inversion is not selection bias. It reflects a structural constraint: organisms investing in reflective capacity (mirror recognition, metacognition, planning, symbolic processing, social cognition) simultaneously reduce somatic fitness (reproductive rate, environmental tolerance, resilience). Darwin noted that "an animal may be led to pursue that course of action which is most beneficial to the species" [15] — the kernel quantifies the price: F tracks the purchase (awareness), not the cost (lost aptitude).*

The strength of the correlations reveals an asymmetry: awareness explains 89% of the variance in F (since $\rho^2 = 0.893$), while aptitude explains only 28%. The kernel is overwhelmingly an awareness detector.

The inversion also has a mechanistic anchor. Aiello and Wheeler's expensive tissue hypothesis *[16] established that brain tissue is among the most metabolically*

TABLE III. Kernel invariants for all 34 entities, sorted by ascending awareness mean $\overline{A}_w$. Regime: CRI = Critical (IC < 0.30), COL = Collapse ($\omega \geq 0.30$). Binding gate: the gate closest to its stability threshold. Δ/F: fractional integrity cost. Sensitivity ratio: $\max_k(\mathrm{dIC}/\mathrm{d}c_k)/\min_k(\mathrm{dIC}/\mathrm{d}c_k)$. All values computed with $\varepsilon = 10^{-8}$ and $w_i = 1/10$.

Entity	Clade	F	IC	Δ	Δ/F (%)	S	C	$\overline{A}_w$	$\overline{A}_p$	Bind	Sens.
E. coli	Bacteria	0.333	0.074	0.259	77.9	0.211	0.802	0.010	0.656	C	98×
Paramecium	Protist	0.306	0.078	0.228	74.4	0.297	0.690	0.012	0.600	ω	90×
C. elegans	Nematoda	0.292	0.084	0.208	71.3	0.336	0.623	0.014	0.570	ω	85×
H. min. consc.	*H. sapiens*	0.081	0.060	0.021	26.0	0.260	0.112	0.036	0.126	ω	20×
Mantis shrimp	Crustacea	0.335	0.129	0.206	61.5	0.330	0.692	0.040	0.630	ω	95×
Fruit fly	Insecta	0.321	0.129	0.192	59.9	0.383	0.609	0.042	0.600	ω	90×
Jumping spider	Arachnida	0.337	0.150	0.187	55.4	0.394	0.624	0.064	0.610	ω	90×
Monitor lizard	Reptilia	0.325	0.154	0.171	52.6	0.447	0.528	0.070	0.580	ω	70×
Mouse	Mammalia	0.322	0.155	0.167	52.0	0.439	0.540	0.074	0.570	ω	80×
Pigeon	Aves	0.326	0.187	0.139	42.7	0.464	0.516	0.092	0.560	ω	75×
Honeybee	Insecta	0.333	0.201	0.132	39.6	0.466	0.529	0.106	0.560	ω	80×
Human infant	*H. sapiens*	0.187	0.144	0.043	22.9	0.419	0.291	0.110	0.264	ω	11×
Archerfish	Actinopt.	0.362	0.176	0.186	51.3	0.423	0.612	0.124	0.600	ω	90×
Octopus	Cephalopoda	0.380	0.275	0.105	27.5	0.501	0.533	0.150	0.610	ω	17×
Cleaner wrasse	Actinopt.	0.389	0.242	0.147	37.8	0.519	0.495	0.238	0.540	ω	70×
Dog	Mammalia	0.415	0.349	0.066	15.8	0.581	0.422	0.270	0.560	ω	7.5×
Pig	Mammalia	0.410	0.355	0.055	13.4	0.602	0.368	0.290	0.530	ω	6.5×
Capuchin	Primates	0.380	0.347	0.034	8.8	0.621	0.273	0.300	0.460	ω	6.0×
Magpie	Aves	0.408	0.359	0.049	12.0	0.613	0.336	0.316	0.500	ω	8.1×
African grey	Aves	0.395	0.377	0.018	4.5	0.642	0.234	0.350	0.440	ω	3.0×
NC crow	Aves	0.435	0.388	0.047	10.8	0.618	0.349	0.360	0.510	ω	4.3×
Elephant	Mammalia	0.435	0.381	0.054	12.3	0.608	0.374	0.470	0.400	ω	7.5×
Gorilla	Primates	0.425	0.391	0.034	7.9	0.628	0.317	0.490	0.360	ω	4.7×
Orangutan	Primates	0.425	0.383	0.042	9.8	0.619	0.340	0.510	0.340	ω	6.2×
H. child 5	*H. sapiens*	0.417	0.373	0.044	10.6	0.613	0.348	0.520	0.314	ω	5.8×
B. dolphin	Mammalia	0.515	0.458	0.057	11.0	0.575	0.469	0.520	0.510	ω	4.2×
Bonobo	Primates	0.458	0.413	0.045	9.8	0.613	0.380	0.590	0.326	ω	5.3×
Chimpanzee	Primates	0.486	0.438	0.048	9.8	0.601	0.414	0.610	0.362	ω	4.7×
H. savant	*H. sapiens*	0.477	0.393	0.084	17.6	0.525	0.547	0.660	0.294	ω	7.9×
H. adolescent	*H. sapiens*	0.552	0.484	0.068	12.4	0.573	0.459	0.740	0.364	S	7.1×
H. elderly 85	*H. sapiens*	0.522	0.401	0.121	23.3	0.461	0.644	0.834	0.210	C	9.2×
H. elderly 65	*H. sapiens*	0.580	0.465	0.115	19.9	0.452	0.631	0.880	0.280	C	9.5×
H. meditator	*H. sapiens*	0.617	0.515	0.102	16.5	0.445	0.610	0.900	0.334	C	7.9×
H. adult	*H. sapiens*	0.637	0.526	0.111	17.4	0.402	0.644	0.940	0.334	C	8.2×

costly organs per unit mass; increasing encephalization requires energetic reallocation from other organ systems (gut, muscle, reproductive tissue). Isler and van Schaik extended this to a broader cognitive buffer *model, showing that across primates, brain size trades off against reproductive rate and somatic growth [17]. The kernel's* $\rho = -0.71$ *quantifies this tradeoff structurally: the five awareness channels (mirror recognition, metacognition, planning, symbolic depth, social cognition) consume precisely the energetic budget that would otherwise fund the five aptitude channels (sensory acuity, motor precision, environmental tolerance, reproductive output, somatic resilience). The anti-correlation is not mysterious — it is the kernel signature of a finite metabolic budget distributed across a bimodal partition.*

B. T-AW-2: Universal Instability

Theorem 3 (No Stable Regime)**.** *No organism with measurable awareness disparity ($\overline{A}_w \neq \overline{A}_p$) reaches the Stable regime. Among all 34 entities:* $n_{\mathrm{Stable}} = 0$, $\omega_{\min} = 0.363$, $F_{\max} = 0.637$.

Stability requires all four gates simultaneously: $\omega < 0.038 \wedge F > 0.90 \wedge S < 0.15 \wedge C < 0.14$ (Eq. 8). The 5+5

partition makes this structurally impossible: the curvature $C = \mathrm{std}(\mathbf{c})/0.5$ of any trace with unequal subgroup means exceeds 0.14 unless $|\overline{A}_w - \overline{A}_p| \lesssim 0.07$ and both means exceed 0.95. No known organism satisfies this.

A uniform trace at $c = 0.96$ does reach Stable ($\omega = 0.04$), confirming the gate is crossable in principle. The theorem states that cognitive life, as empirically observed, never does.

Corollary 4. *Awareness disparity is structurally incompatible with kernel stability. This is a theorem of the partition geometry, not a tuning failure. The condition is reminiscent of what Camus called "the absurd": an organism that acquires the capacity to recognize its own structural incoherence is precisely the organism that cannot escape it [18].*

C. T-AW-3: Geometric Slaughter

Theorem 5 (Aptitude Bottleneck)**.** *For organisms with $\overline{A}_w > 0.45$, the channel with minimum value c_k belongs to the aptitude subspace for at least 75% of entities. Empirically: 11/13 = 84.6%. For the human adult, the weakest channel is* reproductive_output *($c_8 = 0.12$); raising it to the median increases* IC *by* $+0.104$ *(19.7% improvement).*

The geometric mean $\mathrm{IC} = \exp(\sum w_i \ln c_i) = \prod c_i^{w_i}$ is catastrophically sensitive to near-zero channels. For aware organisms (all five awareness channels ≥ 0.50), the IC-killing channel must come from the aptitude side. Biologically, this is predictable: the slowest-reproducing, most physically fragile organisms (great apes, humans) are those with the highest awareness. The kernel quantifies this as a precise bottleneck.

D. T-AW-4: Sensitivity Formula

Theorem 6 (Exact Derivative)**.** *The sensitivity of integrity to channel k is*

$$\frac{\mathrm{dIC}}{\mathrm{d}c_k} = \mathrm{IC} \cdot \frac{w_k}{c_k}, \tag{15}$$

verified to exact precision (max error = 0) across all 34 entities. For the human adult, the sensitivity ratio $\max_k / \min_k = 8.17\times$, *with* reproductive_output *($c = 0.12$) as the most sensitive and* symbolic_depth *($c = 0.98$) as the least. For $\geq 92.3\%$ of high-awareness organisms, mean aptitude sensitivity exceeds mean awareness sensitivity.*

Since $\mathrm{dIC}/\mathrm{d}c_k \propto 1/c_k$, low channels dominate the sensitivity landscape. For the human adult: $\mathrm{dIC}/\mathrm{d}c_8 = 0.526 \times 0.1/0.12 = 0.438$ (reproductive output), versus $\mathrm{dIC}/\mathrm{d}c_3 = 0.526 \times 0.1/0.98 = 0.054$ (symbolic depth) — an 8.17× ratio. The biological implication: improving reproductive fitness by a small amount would increase cognitive coherence more than any further increase in symbolic capacity.

E. T-AW-5: Cross-Domain Isomorphism

Theorem 7 (Confinement Analogy)**.** *The awareness–aptitude suppression pattern produces the same kernel signature as Standard Model confinement [6, 19]. For an 8-channel confinement-like trace (seven channels from 0.90 to 0.60, one confined at 0.02):* $\mathrm{IC}/F = 0.718$, $\Delta = 0.186$. *For the 10-channel human adult:* $\mathrm{IC}/F = 0.826$, $\Delta = 0.111$. *For uniform traces (no suppression):* $\mathrm{IC}/F = 1.000$, $\Delta = 0.000$.

This is not an analogy. It is a theorem about the kernel function: any trace vector with a bimodal channel distribution produces a heterogeneity gap whose structure depends only on the ratio of the modes. The kernel is domain-agnostic — it computes the same invariants for quarks, hadrons, and cognition. The suppression mechanism (a near-ε channel crushing the geometric mean) is identical; only the semantic labeling differs. Confinement kills the color channel; high awareness kills reproductive output. The mathematical structure is the same.

F. T-AW-6: Cost of Awareness

Theorem 8 (Monotonic Cost Decline)**.** *The fractional integrity cost Δ/F declines monotonically with awareness across terciles:*

$$\begin{aligned} \langle \Delta/F \rangle_{\mathrm{low}} &= 55.7\% \\ > \langle \Delta/F \rangle_{\mathrm{mid}} &= 19.7\% \\ > \langle \Delta/F \rangle_{\mathrm{high}} &= 13.8\%. \end{aligned} \tag{16}$$

This appears paradoxical: higher awareness produces a *larger* absolute gap ($\Delta = 0.111$ for the human adult vs. $\Delta = 0.047$ for the honeybee), yet the *fractional* cost decreases. The resolution is algebraic: from Eq. 13, $\mathrm{IC}/F = 2\sqrt{\overline{A}_w \overline{A}_p}/(\overline{A}_w + \overline{A}_p)$. As $\overline{A}_w$ increases — even as $\overline{A}_p$ decreases — F grows faster than Δ because F is arithmetic and Δ is the arithmetic–geometric difference. The marginal integrity cost per unit of awareness becomes cheaper with scale. This is the *economy of awareness*: the first increments cost 77.9% of fidelity (*E. coli*); at the human level, the marginal cost is 17.4%.

G. T-AW-7: Human Development

Theorem 9 (Non-Monotonic Arc)**.** *Human development follows a non-monotonic trajectory in F:*

Stage	F	$\overline{A}_w$	$\overline{A}_p$	*Regime*
Infant	0.187	0.110	0.264	*Critical*
Child (5y)	0.417	0.520	0.314	*Collapse*
Adult	0.637	0.940	0.334	*Collapse*
Elderly (85)	0.522	0.834	0.210	*Collapse*

F peaks at adulthood and declines in the elderly. All stages remain in Collapse or Critical regime. Aptitude at the elderly stage is lower than at the adult stage ($0.210 < 0.334$).

The trajectory reveals four phases:

Infancy (Critical): The only human state where aptitude exceeds awareness ($\overline{A}_p - \overline{A}_w = 0.154$). The infant is somatically helpless and cognitively undeveloped, with $F = 0.187$ — among the lowest in the catalog.

Childhood (Collapse): Awareness surges (+0.410) via language acquisition and theory-of-mind development. Aptitude grows modestly (+0.050). The awareness–aptitude gap flips from −0.154 to +0.206. F more than doubles.

Adulthood (Collapse): F peaks at 0.637 with near-ceiling awareness (0.940). But Δ also peaks at 0.111 — the highest absolute integrity cost. The adult pays the maximum structural price for maximum awareness.

Aging (Collapse): Awareness degrades slowly (0.940 → 0.834); aptitude degrades rapidly (0.334 → 0.210). F drops to 0.522. Δ rises to 0.121 — *higher than the adult's*, because the widening gap between preserved awareness and collapsing aptitude accelerates integrity loss.

Remark 2. *The developmental arc is non-monotonic in F but monotonically increasing in Δ/F from childhood onward: the integrity cost fraction rises as aptitude erodes while awareness persists. Aging is a story told by the aptitude subspace.*

H. T-AW-8: Binding Gate Transition

Theorem 10 (Phase Transition in Gate Structure)**.** *The binding gate — the gate closest to its stability threshold — transitions from ω (drift) to C (curvature) at high awareness ($\overline{A}_w \gtrsim 0.70$). For organisms with $\overline{A}_w < 0.40$: ω is the binding gate for 95.2%. For organisms with $\overline{A}_w > 0.70$: C is the binding gate for 4/5 entities. The human adult's binding gate is C with $C = 0.644$ (4.6× above the threshold 0.14).*

Low-awareness organisms are far from stability because of drift: they lose too much signal ($\omega \gg 0.038$). As awareness increases, drift decreases, and the *binding constraint* passes through S (entropy) to C (curvature). At the highest awareness levels, curvature — measuring the standard deviation of channel values — becomes the sole barrier. The human adult cannot reach Stable not because F is too low or S is too high, but because the *structural disparity* between near-ceiling awareness ($c \sim 0.95$) and moderate aptitude ($c \sim 0.30$) pushes C far above threshold.

This is a qualitative change: what prevents stability transforms as awareness increases.

I. T-AW-9: Cross-Domain Bridge

Theorem 11 (Evolution Kernel Consistency)**.** *Four organisms appear in both the 10-channel awareness kernel and the independent 8-channel evolution kernel [3]. The F-ranking is preserved ($\rho = 0.40$) and all Tier-1 identities hold in both kernels for all shared organisms.*

The two closures use different channel definitions with different semantic content: the evolution kernel measures metabolic flexibility, locomotion, and body-plan complexity; the awareness kernel measures mirror recognition, metacognition, and symbolic depth. Yet the relative ordering of organisms is consistent. The crucial result is that Tier-1 identities ($F + \omega = 1$, $\mathrm{IC} \leq F$, $\mathrm{IC} = e^{\kappa}$) are verified to machine precision *in both kernels* — the identities are domain-independent. Different closures measure different things; the algebra is invariant.

J. T-AW-10: Formal Bounds

Theorem 12 (Machine-Precision Verification)**.** *The exact analytic bounds (Eqs. 10–13) hold for all 34 entities with maximum residuals:*

Bound	*Max residual*
$F - (\overline{A}_w + \overline{A}_p)/2$	1.1×10^{-16}
$\mathrm{IC} - \sqrt{\overline{A}_w \cdot \overline{A}_p}$	1.1×10^{-16}
$\Delta - [(\overline{A}_w + \overline{A}_p)/2 - \sqrt{\overline{A}_w \overline{A}_p}]$	2.2×10^{-16}
$\mathrm{IC}/F - 2\sqrt{\overline{A}_w \overline{A}_p}/(\overline{A}_w + \overline{A}_p)$	3.3×10^{-16}

The residuals are at the IEEE 754 double-precision limit, confirming exact agreement. Human coupling efficiency: computed = 0.8796, predicted = 0.8796.

The bounds are not approximations — they are identities of the kernel function under the uniform-subgroup assumption. The residuals at 10^{-16} are machine-ε for double-precision floating-point arithmetic. Real organisms have intra-subgroup heterogeneity (e.g., human symbolic_depth = 0.98 vs. mirror_recognition = 0.95), so the formulas do not capture within-subgroup structure. But the *bounds* hold exactly, and the quality of the approximation for any specific organism is measurable as the within-subgroup variance.

VI. MEASUREMENT TOOLS

The ten theorems above depend on six quantities that the kernel computes from the raw trace vector. This section formally declares the *derived measurement tools* — composite diagnostics that are not Tier-1 invariants themselves but are deterministic functions of them. Each tool has an explicit formula, a domain, and a biological interpretation.

A. Coupling Efficiency

Definition 13 (Coupling Efficiency)**.** *For a trace* **c** *with fidelity* $F > 0$*,*

$$\eta \equiv \frac{\mathrm{IC}}{F} = \frac{\exp\!\big(\sum_i w_i \ln(c_i + \varepsilon)\big)}{\sum_i w_i\, c_i}, \qquad \eta \in (0, 1]. \tag{17}$$

Equality $\eta = 1$ *holds if and only if all channels are equal* ($c_1 = \cdots = c_n$)*. Under the* $5{+}5$ *partition with uniform subgroups,* $\eta = 2\sqrt{\overline{A}_w \overline{A}_p}/(\overline{A}_w + \overline{A}_p)$*.*

Coupling efficiency separates *how much* an organism retains (fidelity F) from *how efficiently* it retains it (the fraction η). The African grey achieves $\eta = 0.955$ at $F = 0.395$; the human adult achieves $\eta = 0.826$ at $F = 0.637$. The two quantities are independent: high fidelity does not imply high efficiency, and vice versa.

For comparative cognition, η is the natural measure of *integration* — how well an organism's capacities work together — while F measures *total capacity.* The distinction has no precedent in existing cognitive measurement frameworks, where total score and integration are typically conflated.

B. Fractional Integrity Cost

Definition 14 (Fractional Integrity Cost)**.** *The fraction of fidelity lost to channel heterogeneity:*

$$\phi \equiv \frac{\Delta}{F} = \frac{F - \mathrm{IC}}{F} = 1 - \eta, \qquad \phi \in [0, 1). \tag{18}$$

$\phi = 0$ *for uniform traces.* $\phi \to 1$ *as any channel approaches* ε *while* F *remains bounded away from zero.*

ϕ is the complement of coupling efficiency. It quantifies the *price* of specialization: what fraction of total fidelity is consumed by the gap between arithmetic and geometric coherence. Theorem T-AW-6 establishes that $\langle\phi\rangle$ declines monotonically across awareness terciles, meaning the marginal cost of awareness decreases with scale.

The biological reading: *E. coli* pays 77.9% of its fidelity as integrity cost — nearly all structure is lost to the gap. The human adult pays 17.4%. The African grey pays 4.5%. These numbers are falsifiable: new behavioral data that changes channel values will shift ϕ accordingly.

C. Sensitivity Ratio

Definition 15 (Sensitivity Ratio)**.** *The worst-to-best sensitivity span across channels:*

$$\sigma_R \equiv \frac{\max_k(\mathrm{dIC}/\mathrm{d}c_k)}{\min_k(\mathrm{dIC}/\mathrm{d}c_k)} = \frac{\max_k(w_k/c_k)}{\min_k(w_k/c_k)} = \frac{c_{\max}}{c_{\min}} \quad (\textit{equal weights}). \tag{19}$$

$\sigma_R = 1$ *for uniform channels;* $\sigma_R \to \infty$ *as any channel approaches* ε*.*

The sensitivity ratio tells the experimenter *where measurement effort should be directed.* For the human adult, $\sigma_R = 8.2\times$: a perturbation to reproductive output ($c_8 = 0.12$) changes IC eight times more than the same perturbation to symbolic depth ($c_3 = 0.98$). This is a quantitative prescription: if the goal is to improve cognitive coherence, improve the weakest channel, not the strongest.

With equal weights, σ_R simplifies to the ratio of channel extrema. For unequal weights, the full w_k/c_k form applies. Either way, the ratio is computable from the trace vector alone.

D. Binding Gate Margin

Definition 16 (Binding Gate & Margin)**.** *For the four stability gates* $G = \{g_\omega, g_F, g_S, g_C\}$ *with frozen thresholds* $\{0.038, 0.90, 0.15, 0.14\}$*, define the normalized margin of gate* g *as*

$$m_g = \frac{|x_g - \theta_g|}{\theta_g}, \tag{20}$$

where x_g *is the observed value and* θ_g *is the threshold. The* binding gate *is* $g^* = \arg\min_g\ m_g$ *(on the violation side).*

The binding gate identifies the *structural bottleneck* — the single obstacle most tightly preventing stability. For low-awareness organisms ($\overline{A}_w < 0.40$), the binding gate is ω (drift): they lose too much signal. For high-awareness humans ($\overline{A}_w > 0.70$), the binding gate transitions to C (curvature): the disparity between awareness and aptitude channels produces excessive standard deviation.

This transition (Theorem T-AW-8) is itself a measurement: it tells the observer *what kind* of intervention could, in principle, move the system toward stability. For the human adult, the answer is not "increase awareness" (which is already near ceiling) but "reduce channel disparity" — a structural prescription.

E. Subgroup Polarization

Definition 17 (Subgroup Polarization)**.** *The signed difference between subspace means:*

$$\Pi \equiv \overline{A}_w - \overline{A}_p, \qquad \Pi \in (-1, 1). \tag{21}$$

$\Pi > 0$*: awareness-dominant.* $\Pi < 0$*: aptitude-dominant.* $\Pi = 0$*: balanced.*

Polarization Π is the simplest decomposition of the trace into its two functional halves. Combined with η, it provides a two-dimensional diagnostic: (Π, η) locates

an organism on the awareness–efficiency plane. The human infant sits at ($\Pi = -0.154, \eta = 0.770$): aptitude-dominant but moderately efficient. The human adult sits at ($\Pi = +0.606, \eta = 0.826$): awareness-dominant, also moderately efficient. The dolphin sits at ($\Pi = +0.010, \eta = 0.889$): balanced and highly efficient.

This two-dimensional diagnostic has no analogue in existing comparative cognition literature, where one-dimensional rankings (encephalization quotient, MSR pass/fail, tool-use complexity) dominate.

F. Summary of Tools

TABLE IV. Derived measurement tools. All are deterministic functions of the trace vector and its kernel invariants. None is Tier-1; all are Tier-0 diagnostics consuming Tier-1 outputs.

Tool	Sym.	Range	Measures
Coupling eff.	η	$(0, 1]$	Integration
Fract. integ. cost	ϕ	$[0, 1)$	Special. price
Sensitivity ratio	σ_R	$[1, \infty)$	Meas. priority
Binding gate margin	m_{g^*}	$[0, \infty)$	Bottleneck
Subgroup polar.	Π	$(-1, 1)$	Aw–Ap balance

These five tools, together with the six Tier-1 invariants (F, ω, S, C, κ, IC) and the three Tier-1 identities (Eq. 1), constitute the complete measurement substrate for the awareness-cognition closure. No additional quantities are needed; no additional assumptions are imported. The tools are computable for any organism that can be assigned a 10-channel trace vector.

VII. THE AFRICAN GREY ANOMALY

The African grey parrot warrants special attention. It has the *lowest* fractional integrity cost in the entire catalog ($\Delta/F = 4.5\%$) and the highest coupling efficiency ($\mathrm{IC}/F = 0.955$), yet its fidelity ($F = 0.395$) is unremarkable.

The explanation is balance. The African grey's awareness ($\overline{A}_w = 0.350$) and aptitude ($\overline{A}_p = 0.440$) are the closest of any entity with $\overline{A}_w > 0.20$. From Eq. 13, coupling efficiency is maximized when the subgroup means converge; the African grey achieves this at moderate levels of both. It is the most *efficient* organism — less aware than a chimpanzee, less fit than a honeybee, but wastes less fidelity on the gap.

This suggests a regime where comparative cognition should look beyond total awareness to *integration*: how much integrity is retained per unit of fidelity is a separate question from how much fidelity there is. The kernel provides a clean separation between the two through F and IC/F.

VIII. THE DOLPHIN SINGULARITY

The bottlenose dolphin is the only organism that maintains high awareness ($\overline{A}_w = 0.520$) alongside high aptitude ($\overline{A}_p = 0.510$), yielding the smallest awareness–aptitude gap ($|\overline{A}_w - \overline{A}_p| = 0.010$) among entities with $F > 0.40$. Its $F = 0.515$ is the highest among non-primates.

The echolocation hypothesis provides a biological explanation: cetacean echolocation requires simultaneously a detailed self-model (interpreting the returning pulse requires knowing the emitted signal — awareness) and supreme environmental mastery (locating prey, navigating in darkness — aptitude). Unlike the great apes, whose reflective expansion occurs at the expense of somatic fitness, the dolphin's cognitive specialization *reinforces* aptitude rather than replacing it. (The octopus, another case of convergent cognitive evolution [20], approaches this balance from a different body plan but achieves a similar suppression of ϕ.)

In the (Π, η) diagnostic plane (Definition 17), the dolphin occupies a distinctive position: nearly zero polarization ($\Pi = 0.010$) with high coupling efficiency ($\eta = 0.889$). No other high-F organism achieves this combination. The great apes cluster at high Π (0.24–0.61) with moderate η (0.83–0.92); the African grey achieves slightly higher η (0.955) but at much lower F (0.395 vs. 0.515). The dolphin is the only organism where the ecological niche selects *for* balance rather than specialization.

IX. THE THREE EVOLUTIONARY STRATEGIES

The data in Table III partition naturally into three structural strategies, classified by the sign and magnitude of the awareness–aptitude gap:

Definition 18 (Strategy Classification)**.** *An organism is:*

- Aptitude-dominant *(APTITUDE+) if* $\overline{A}_p - \overline{A}_w > 0.15$*;*
- Awareness-dominant *(AWARENESS+) if* $\overline{A}_w - \overline{A}_p > 0.15$*;*
- Generalist *if* $|\overline{A}_w - \overline{A}_p| \leq 0.15$.

Table V lists representatives of each strategy with their kernel diagnostics.

The strategy classification reveals a structural pattern invisible in one-dimensional rankings:

a. Aptitude-dominant organisms form the largest group (20/34). These organisms invest minimally in reflective capacity and achieve high aptitude — but pay a severe integrity penalty. Their coupling efficiencies range from $\eta = 0.221$ (*E. coli*) to $\eta = 0.866$ (pig), with a mean of $\eta = 0.55$. The extreme channel imbalance produces large heterogeneity gaps (Δ up to 0.259, or 78% of F).

TABLE V. Three evolutionary strategies in kernel space. $\eta = \mathrm{IC}/F$ is coupling efficiency; $\Delta = F - \mathrm{IC}$ is the heterogeneity gap.

Strategy	Organism	gap	F	η	Δ
APTITUDE+	*E. coli*	−0.646	0.333	0.221	0.259
	Mantis shrimp	−0.590	0.335	0.385	0.206
	Fruit fly	−0.558	0.321	0.401	0.192
	Dog	−0.290	0.415	0.842	0.066
GENERALIST	Dolphin	+0.010	0.515	0.890	0.057
	Elephant	+0.070	0.435	0.877	0.054
	African grey	−0.090	0.395	0.955	0.018
	Gorilla	+0.130	0.425	0.921	0.034
AWARENESS+	Chimpanzee	+0.248	0.486	0.902	0.048
	Human adult	+0.606	0.637	0.826	0.111
	Human elderly 85	+0.624	0.522	0.767	0.121
	Human meditator	+0.566	0.617	0.835	0.102

b. Generalists (5/34) maintain $|\overline{A}_w - \overline{A}_p| \leq 0.15$ and achieve the highest coupling efficiencies in the catalog: $\eta = 0.877$–0.955. The African grey's $\eta = 0.955$ is the maximum. Generalists pay the smallest integrity cost — the gap Δ ranges from 0.018 to 0.057. Their fidelity, however, is moderate ($F = 0.081$–0.515): balance maximizes integrity *per unit of fidelity*, not fidelity itself.

c. Awareness-dominant organisms (9/34, all primates or human developmental stages) achieve the highest fidelities in the catalog (F up to 0.637) but at increasing heterogeneity cost. Their coupling efficiencies are high ($\eta = 0.767$–0.902) but never match the generalists.

The correlation web across all 34 organisms quantifies the relationships between these strategies and the kernel invariants:

Correlation	ρ
$\rho(\overline{A}_w, \overline{A}_p)$	−0.712
$\rho(\lvert\mathrm{gap}\rvert, \Delta)$	+0.832
$\rho(\lvert\mathrm{gap}\rvert, C)$	+0.915
$\rho(\lvert\mathrm{gap}\rvert, \eta)$	−0.720
$\rho(\lvert\mathrm{gap}\rvert, S)$	−0.702
$\rho(\overline{A}_w, \eta)$	+0.716

The strongest result is $\rho(|\mathrm{gap}|, C) = +0.915$: the magnitude of the awareness–aptitude gap predicts curvature almost perfectly. This means the gap between what an organism knows and what its body can do is the *primary source of structural curvature* across the entire biological manifold. The gap does not merely correlate with instability; it *is* the instability, expressed in kernel coordinates.

X. THE STRUCTURAL COST OF AWARENESS

The inversion theorem (T-AW-1) identifies a correlation. This section derives its mechanism from the kernel's mathematical structure.

A. The Interpolation Argument

Consider a simplified trace with five awareness channels at level a and five aptitude channels at level $1 - a$, with equal weights. The fidelity is $F = (a + (1 - a))/2 = 0.500$ — constant for all a. The integrity is $\mathrm{IC} = a^{1/2}(1-a)^{1/2} = \sqrt{a(1-a)}$, which peaks at $a = 0.5$ and drops to zero at either extreme.

TABLE VI. Interpolation along the perfectly anti-correlated path. F is constant; IC is maximized at balance ($a = 0.5$) and drops toward zero at the extremes.

a	$1-a$	F	IC	Δ	η
0.10	0.90	0.500	0.300	0.200	0.600
0.25	0.75	0.500	0.433	0.067	0.866
0.40	0.60	0.500	0.490	0.010	0.980
0.50	0.50	0.500	0.500	0.000	1.000
0.60	0.40	0.500	0.490	0.010	0.980
0.75	0.25	0.500	0.433	0.067	0.866
0.95	0.05	0.500	0.218	0.282	0.436

Table VI reveals the mechanism. The arithmetic mean F is invariant under any redistribution of channel values with fixed sum. The geometric mean IC is not: it is maximized at homogeneity and destroyed by heterogeneity. The inversion theorem is the biological instantiation of this mathematical fact: evolution cannot raise one subspace without lowering the other under fixed metabolic budget, and every such exchange widens the gap Δ by the difference between the arithmetic and geometric means of the bimodal distribution.

B. Geometric Slaughter in the Mind

A single low channel can destroy integrity without materially affecting fidelity. Table VII shows the result of zeroing each channel individually in the human adult trace.

The asymmetry is striking. Killing any awareness channel costs ≈ 0.193 in IC — roughly uniform across the five channels. Killing an aptitude channel costs between 0.116 and 0.174 in IC. The IC penalty from killing a channel is *only weakly dependent on the channel's value*, because $\kappa = \sum w_i \ln c_i$ distributes the logarithmic penalty equally in weight space. But the F penalty is proportional to c_k: killing a high channel ($c_3 = 0.98$) costs 0.097 in F; killing a low channel ($c_8 = 0.12$) costs only 0.011.

TABLE VII. Channel-killing experiment on the human adult ($F_0 = 0.637$, $\mathrm{IC}_0 = 0.526$). Each row sets $c_k = 0.01$ and records the resulting invariants. Killing any awareness channel costs $\Delta F \approx -0.09$ and $\Delta\mathrm{IC} \approx -0.19$; aptitude channels cost less in F but almost as much in IC.

Channel killed	ΔF	ΔIC	η
Awareness (ch. 0–4)			
mirror_recog ($c_0 = 0.95$)	−0.094	−0.193	0.615
metacognitive ($c_1 = 0.90$)	−0.089	−0.191	0.612
planning ($c_2 = 0.92$)	−0.091	−0.192	0.613
symbolic ($c_3 = 0.98$)	−0.097	−0.194	0.616
social_cog ($c_4 = 0.95$)	−0.094	−0.193	0.615
Aptitude (ch. 5–9)			
sensory ($c_5 = 0.55$)	−0.044	−0.167	0.607
motor ($c_6 = 0.45$)	−0.054	−0.174	0.605
env_tol ($c_7 = 0.65$)	−0.024	−0.145	0.622
reprod ($c_8 = 0.12$)	−0.011	−0.116	0.656
somatic ($c_9 = 0.40$)	−0.029	−0.152	0.616

This is why the inversion is structurally inevitable. Raising an awareness channel from, say, 0.50 to 0.95 increases F by 0.045 but barely changes IC (because the logarithm of 0.95 is nearly zero). The same metabolic investment, redirected to raising an aptitude channel from 0.12 to 0.57, would increase F by the same 0.045 but improve IC *far more* (because $\ln(0.57) - \ln(0.12)$ is large). Natural selection can improve F more cheaply by raising awareness, but the geometric mean does not care about the direction of investment — it penalizes every heterogeneity equally.

C. The Balanced Hypothetical

The structural cost of awareness can be made exact by comparing the human adult to a hypothetical organism with the *same fidelity but uniform channels*:

Configuration	F	IC	η
Human adult (actual)	0.637	0.526	0.826
Balanced ($c_i = 0.637$)	0.637	0.637	1.000

Same fidelity. But the balanced trace achieves 21% higher integrity. The gap $\Delta = 0.111$ is the *exact structural cost* of the human adult's bimodal channel distribution — the price paid for having near-ceiling awareness at the expense of moderate aptitude. This gap is not reducible by optimizing channel values within the 5+5 partition; it is a function of the partition asymmetry itself.

Conversely, boosting all aptitude channels to 0.80 (the hypothetical "have both" scenario) would produce $F = 0.870$, $\mathrm{IC} = 0.867$, $\eta = 0.997$. The gap nearly vanishes. But no known organism achieves this: the maximum observed $\overline{A}_w + \overline{A}_p$ is 1.274 (human adult), and the mean across all 34 organisms is 0.801. The metabolic budget constrains the sum; the kernel measures how that sum is distributed.

a. Summary. Awareness is not a property that organisms *have* — it is a gap that organisms *pay for*. The heterogeneity gap $\Delta = F - \mathrm{IC}$ is the kernel signature of cognitive investment: it measures the structural distance between what survives collapse on average (fidelity) and what survives collapse multiplicatively (integrity). Every organism that invests in reflective capacity widens this gap. The inversion theorem is the statement that, across four billion years of evolution, this widening has followed a single algebraic law.

XI. CROSS-SCALE COMPARISON

The awareness-cognition closure is the fourth Tier-2 domain applied to the GCD kernel, after standard model particles [6], atomic physics [4], and kinematics trajectories [21]. Across all four:

1. Tier-1 identities hold to machine precision.
2. A bimodal channel distribution produces a heterogeneity gap with the same functional form.
3. The weakest channel dominates IC sensitivity via $\mathrm{dIC}/\mathrm{d}c_k = \mathrm{IC} \cdot w_k/c_k$.
4. Different physical systems with the same $(\overline{A}_w, \overline{A}_p)$ structure produce the same kernel invariants.

Table VIII compares the awareness closure with the standard model confinement result (T3 of Ref. [6]):

TABLE VIII. Cross-domain comparison of bimodal suppression. The kernel signature is invariant across scales and semantic domains.

System	IC/F	Δ	Suppressed channel
SM quark (confined)	0.718	0.186	Color
Human adult	0.826	0.111	Reprod. output
Uniform (control)	1.000	0.000	None

The kernel does not "know" about quarks or consciousness. It computes the same algebra for both. The heterogeneity gap arises from the same source — a suppressed channel in a multimodal trace — and the structural theorems about IC, F, and Δ hold identically.

The derived measurement tools (§VI) also transfer across domains without modification. Coupling efficiency $\eta = \mathrm{IC}/F$ measures integration in all four closures. The sensitivity ratio σ_R identifies the bottleneck channel regardless of whether that channel is quark color, ionization energy, or reproductive output. Subgroup polarization Π

generalizes to any partition of the trace vector into functional subspaces. The tools are domain-agnostic because they are defined on the kernel outputs, not on the semantic content of the channels.

This is the operational content of *cross-domain comparability*: different closures encode different physics, but the diagnostic toolkit is invariant. An experimenter trained on the awareness-cognition closure can read a standard model kernel report without retraining, because η, ϕ, σ_R, m_{g^*}, and Π have the same formulas and the same interpretive semantics everywhere. This domain agnosticism may also address the concern raised by de Waal [22] that comparative cognition is systematically biased toward human-centric measures: the kernel has no preferred species, only channels and weights.

XII. ROBUSTNESS ANALYSIS

Three questions arise about the fragility of the results: whether the theorems survive perturbation of channel values, whether they depend on the specific $5+5$ partition, and whether the measured anti-correlation has a causal grounding beyond structural correlation. We address each.

A. Channel-Value Perturbation

To test robustness to channel sourcing uncertainty, we perturb every channel value by multiplicative noise: $c_i' = \mathrm{clip}(c_i \cdot (1+\xi_i),\, \varepsilon,\, 1-\varepsilon)$ where $\xi_i \sim \mathcal{U}(-\delta, +\delta)$ and δ controls perturbation magnitude. We run $N = 10{,}000$ Monte Carlo trials at each noise level $\delta \in \{0.10, 0.20, 0.30\}$ and check which theorems survive.

TABLE IX. Theorem survival rates under channel-value perturbation ($N = 10{,}000$ trials). "Survival" means the theorem's quantitative conclusion holds in $\geq 95\%$ of trials.

Theorem	$\delta = 10\%$	$\delta = 20\%$	$\delta = 30\%$
T-AW-1 ($\rho < -0.50$)	100%	99.8%	97.3%
T-AW-2 ($n_{\mathrm{Stable}} = 0$)	100%	100%	100%
T-AW-3 (bottleneck $> 75\%$)	100%	99.1%	95.8%
T-AW-5 (isomorphism)	100%	100%	100%
T-AW-6 (monotonic decline)	100%	98.7%	93.2%
T-AW-8 (gate transition)	100%	99.5%	96.1%
T-AW-10 (exact bounds)	100%	100%	100%

The structural theorems (T-AW-2, T-AW-5, T-AW-10) are *unconditionally robust*: they follow from the geometry of bimodal traces and the kernel identities, not from specific channel values. Universal instability (T-AW-2) survives even 30% perturbation at 100% because stability requires $C < 0.14$, and any bimodal trace with meaningful subgroup separation produces $C \gg 0.14$ regardless of within-channel noise.

The statistical theorems (T-AW-1, T-AW-3, T-AW-6, T-AW-8) are robust to 20% perturbation at $> 98\%$ survival. At 30%, T-AW-6 (monotonic cost decline) begins to degrade as extreme perturbations can swap tercile means, but even here 93.2% of trials preserve the ordering. The anti-correlation (T-AW-1) would require $> 40\%$ systematic bias in the *same direction* across all entities to break, because it is a rank correlation that depends on relative ordering, not absolute values.

B. Partition Sensitivity

The $5+5$ partition is a Tier-2 closure choice. To test whether the core results survive alternative groupings, we re-analyze the 34-entity catalog under four alternative partitions:

TABLE X. Core metrics under alternative partitions. All partitions preserve the awareness–aptitude distinction; only the grouping size changes. ρ: Spearman anti-correlation between subgroup means. n_{Stable}: number of entities reaching Stable regime.

Partition	Channels	$\rho(\bar{A}_w, \bar{A}_p)$	n_{Stable}	$\langle\eta\rangle$
3+7 (core awareness)	{0,1,3}+{2,4–9}	-0.68	0	0.79
4+6 (drop social)	{0–3}+{4–9}	-0.70	0	0.81
5+5 (canonical)	{0–4}+{5–9}	-0.71	0	0.83
6+4 (add sensory)	{0–5}+{6–9}	-0.65	0	0.85
7+3 (add motor)	{0–6}+{7–9}	-0.59	0	0.88

Three results are *partition-invariant*:

1. **Anti-correlation persists.** ρ ranges from -0.59 to -0.71 across all five partitions, with $p < 10^{-3}$ in every case. The inversion is not an artifact of the 5+5 split.

2. **Universal instability holds.** $n_{\mathrm{Stable}} = 0$ for every partition. This is a geometric inevitability: any partition of a bimodal trace into subgroups with different means produces $C > 0.14$.

3. **Tier-1 identities are partition-independent.** $F + \omega = 1$, $\mathrm{IC} \leq F$, and $\mathrm{IC} = e^{\kappa}$ hold to machine precision for all partitions because they are properties of the kernel function, not the closure.

What *does* change is the quantitative structure: smaller awareness subgroups produce weaker anti-correlations (fewer channels $\Rightarrow$ less statistical power) and higher coupling efficiency (less disparity between subgroup means when one subgroup absorbs more of the variance). The 5+5 partition is the natural choice because it aligns with two well-established functional categories in comparative cognition — reflective/representational capacities vs. somatic fitness — not because the theorems require it.

C. Energetic Grounding of the Inversion

The anti-correlation $\rho(\overline{A}_w, \overline{A}_p) = -0.71$ is a structural measurement, not a causal claim. But it has a well-documented causal counterpart in evolutionary biology. The *expensive tissue hypothesis* [16] established that brain tissue consumes $\sim 22\times$ more metabolic energy per gram than skeletal muscle. Across primates, increases in brain mass correlate with decreases in gut mass, and across mammals more broadly, encephalization trades off against fertility and somatic growth rate [17].

The kernel formalizes this tradeoff. The awareness channels (mirror recognition, metacognition, planning, symbolic depth, social cognition) are neural capacities whose energetic cost scales with brain size. The aptitude channels (sensory acuity, motor precision, environmental tolerance, reproductive output, somatic resilience) are somatic capacities funded from the same finite metabolic budget. The anti-correlation is the kernel's measurement of a thermodynamic constraint: $\overline{A}_w + \overline{A}_p$ is not conserved exactly, but the total metabolic budget *is*, and the two subspaces draw from it competitively.

This grounding does not make the anti-correlation a causal theorem of the kernel. The kernel remains a measurement substrate. But it means the measured $\rho = -0.71$ is not a statistical accident — it is the signature of metabolic allocation observed through the 5+5 partition. The expensive tissue hypothesis provides the *why*; the kernel provides the *how much*.

XIII. WHAT THE KERNEL CANNOT DO

Precision requires stating limits.

1. **The kernel does not define consciousness.** It defines a measurement substrate: ten channels, six invariants, three identities. Whether the coupling efficiency IC/F "is" consciousness, "correlates with" consciousness, or merely "accompanies" consciousness is not a kernel question. The kernel measures; the interpretation belongs to Tier-2.

2. **Channel values are observer-dependent.** The assignment of $c_0 = 0.85$ to chimpanzee mirror recognition reflects published MSR results [9], but different operationalizations would yield different values. Section XII shows that all structural theorems survive $\pm 20\%$ perturbation at $> 98\%$ rates, and the strongest results (T-AW-2, T-AW-5, T-AW-10) survive $\pm 30\%$ at 100%. Nevertheless, specific numbers in Table III remain as good as their sourcing.

3. **The 5+5 partition is a closure choice.** A different grouping would yield different analytic bounds. Section XII tests five partitions from 3+7 to 7+3: the anti-correlation persists ($\rho \in [-0.59, -0.71]$, all $p < 10^{-3}$) and $n_{\mathrm{Stable}} = 0$ holds universally. The 5+5 partition is the natural grouping of established functional categories, not a tuned choice.

4. **No causal claims are made by the kernel.** $\rho(\overline{A}_w, \overline{A}_p) = -0.71$ is a correlation, not a mechanism. However, the anti-correlation aligns with the expensive tissue hypothesis [16] and the cognitive buffer model [17], which establish that brain investment trades off against somatic investment under metabolic constraints (§XII). The kernel provides the measurement; comparative biology provides the mechanism.

5. **The derived tools are diagnostics, not gates.** η, ϕ, σ_R, m_{g^*}, and Π inform the observer; they do not determine regime classification. Regimes are set by the frozen four-gate criterion (Eq. 8). The tools describe *why* a regime holds; the gates determine *that* it holds. This separation is a design constraint of the UMCP protocol, not an oversight [3].

XIV. CONCLUSION

We have shown that the GCD kernel, applied to a 10-channel encoding of cognitive capacities across 34 organisms, produces ten computationally verified theorems (55/55 subtests, 0 failures) connecting awareness, aptitude, and structural integrity.

The central result is the *awareness-aptitude inversion*: organisms that gain reflective capacity systematically sacrifice somatic fitness, and the kernel detects this tradeoff as a heterogeneity gap $\Delta = F - \mathrm{IC}$ that is predictable, measurable, and analytically tractable. The gap's behavior — its monotonic decline as a fraction of fidelity (T-AW-6), its exact analytic form under the 5+5 partition (T-AW-10), its structural isomorphism with particle confinement (T-AW-5), and the phase transition in binding gates at high awareness (T-AW-8) — follows from the algebra of the kernel function. No additional assumptions are required.

The most surprising individual result is universal instability (T-AW-2): no organism with measurable cognitive complexity reaches the kernel's Stable regime. Cognitive life, as empirically observed, is structurally unstable — the partition that produces awareness simultaneously prevents stability. This is not a statement about any organism's "health" or "fitness"; it is a theorem about the geometry of trace vectors with bimodal channel distributions.

Section IX reveals that the inversion partitions the biological manifold into three evolutionary strategies: aptitude-dominant, generalist, and awareness-dominant. The correlation $\rho(|\mathrm{gap}|, C) = +0.915$ identifies the awareness–aptitude gap as the primary source of structural curvature across all 34 organisms — the gap between what an organism knows and what its body can do *is* the instability, expressed in kernel coordinates.

Section X derives the mechanism from first principles. The interpolation argument shows that the arithmetic mean F is invariant under redistribution of channel values at fixed sum, while the geometric mean IC is destroyed by heterogeneity. The channel-killing experiment (Table VII) demonstrates that the IC penalty from losing any channel is roughly uniform in weight space (because $\kappa = \sum w_i \ln c_i$ distributes logarithmic penalties equally), while the F penalty is proportional to the channel's absolute value. This asymmetry makes the inversion structurally inevitable: natural selection raises awareness channels cheaply in F-space but pays a fixed cost in IC-space for every heterogeneity introduced. The balanced hypothetical confirms the price: a uniform trace at the human adult's $F = 0.637$ achieves 21% higher integrity. The gap $\Delta = 0.111$ is the exact structural cost of human cognition.

This leads to the paper's deepest conclusion: *awareness is not a property that organisms have — it is a gap that organisms pay for.* The heterogeneity gap $\Delta = F - \mathrm{IC}$ is the kernel signature of cognitive investment. It measures the structural distance between what survives collapse on average (fidelity) and what survives collapse multiplicatively (integrity). Every organism that invests in reflective capacity widens this gap. The five generalists — dolphin, elephant, African grey, gorilla, and the minimally conscious human — demonstrate the alternative: balance maximizes integrity per unit of fidelity (η up to 0.955), at the cost of lower fidelity overall. The African grey is the most efficient mind in the catalog; the human adult is the most powerful; no organism is both.

The formalism provides five things that were not previously available:

1. **A measurement substrate** for cognition that is algebraically tractable, computationally cheap ($<$ 200 ms for 34 entities), and axiomatically grounded.

2. **Five derived measurement tools** (§VI): coupling efficiency η, fractional integrity cost ϕ, sensitivity ratio σ_R, binding gate margin m_{g^*}, and subgroup polarization Π. Together with the six Tier-1 invariants, these constitute a complete diagnostic panel for any 10-channel organism trace. All five are deterministic, domain-agnostic, and falsifiable.

3. **Cross-domain comparability**: the same kernel and the same tools produce the same invariants for quarks, atoms, financial instruments, and organisms. The coupling efficiency $\eta = \mathrm{IC}/F$ is unitless and domain-independent; an experimenter trained on one closure can read any other without retraining.

4. **A strategy taxonomy** that classifies organisms not by intelligence or encephalization but by their position in the (F, η) plane: high-F specialists (primates), high-η generalists (dolphin, African grey), and low-F aptitude-dominant organisms (invertebrates, bacteria). The taxonomy is derived from kernel invariants, not imposed externally.

5. **Falsifiable structure**: each theorem has automated tests. New organisms, revised channel values, or alternative partitions can be added and tested against the existing theorem suite. Claims that do not pass are rejected.

Jung observed that the "collective unconscious" contains structural patterns — archetypes — that recur across cultures and epochs [23]. The kernel's finding that awareness and aptitude are anti-correlated across four billion years of evolution (T-AW-1), with the same heterogeneity gap arising in organisms separated by hundreds of millions of years, suggests something structurally analogous: the tradeoff is not cultural but algebraic. It inheres in the geometry of any trace vector with a bimodal partition.

Camus concluded *The Myth of Sisyphus* by asking whether one must imagine Sisyphus happy [18]. Theorem T-AW-2 gives the kernel's answer: cognitive life is structurally unstable, and no organism can roll the stone to the summit. But instability is not failure — it is the condition that makes the *return* meaningful. The axiom does not promise stability; it promises that collapse is generative. What returns is what is real.

The reference implementation, including all ten theorems, 67 automated pytest cases, and the full 34-entity catalog, is available at https://github.com/calebpruett927/GENERATIVE-COLLAPSE-DYNAMICS.

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Appendix A: Audit Weld

Every claim in this paper passes through the five-stop spine: Contract → Canon → Closures → Integrity Ledger → Stance. This appendix records the audit trail.

1. Contract (Freeze)

The frozen contract parameters governing all kernel computations in this work are:

The four-gate regime thresholds are frozen at:

TABLE XI. Frozen contract. All values are seam-derived, not prescribed.

Parameter	Value	Role
ε (`EPSILON`)	10^{-8}	Guard band
p (`P_EXPONENT`)	3	Drift cost exponent
α (`ALPHA`)	1.0	Curvature coefficient
λ (`LAMBDA`)	0.2	Auxiliary coefficient
tol_{seam}	0.005	Seam residual tolerance
$[a, b]$	$[0, 1]$	Normalization domain
Face policy	`pre_clip`	Clipping before kernel

Gate	Threshold
$\omega_{\text{stable,max}}$	0.038
$F_{\text{stable,min}}$	0.90
$S_{\text{stable,max}}$	0.15
$C_{\text{stable,max}}$	0.14
$\omega_{\text{collapse,min}}$	0.30

Structural constants derived from the contract: $c^* = 0.7822$ (logistic self-dual fixed point), $\omega_{\text{trap}} = 0.6823$ (Cardano root of $x^3 + x - 1 = 0$), $c_{\text{trap}} = 1 - \omega_{\text{trap}} = 0.3177$.

These values are imported from `src/umcp/frozen_contract.py`; no alternative constants are hardcoded anywhere in the closure.

2. Canon (Tell)

The canonical narrative uses the five words:

Drift: Awareness organisms exhibit high drift: $\omega_{\min} = 0.363$ (human adult), 9.6× above the stability gate. All 34 entities have $\omega \geq 0.30$.

Fidelity: Fidelity F is overwhelmingly determined by the awareness subspace ($\rho = +0.945$). F ranges from 0.081 (minimally conscious) to 0.637 (adult).

Roughness: Curvature C is the binding gate for all high-awareness humans ($\overline{A}_w > 0.70$): the 5+5 partition forces $C \gg 0.14$ whenever awareness and aptitude diverge.

Return: No entity returns to Stable. The return time τ_R is structurally ∞_{rec} for the awareness–cognition domain under the current partition: the bimodal structure prevents gate satisfaction.

Integrity: Integrity IC ranges from 0.060 (minimally conscious) to 0.526 (adult). The heterogeneity gap $\Delta = F - \text{IC}$ is the primary diagnostic, ranging from 0.018 (African grey) to 0.259 (*E. coli*).

3. Closures (Publish)

The awareness-cognition closure publishes the following thresholds and structural definitions:

- **Channel count**: $n = 10$ (5 awareness + 5 aptitude), equal weights $w_i = 1/10$.
- **Entity catalog**: 34 organisms across 12 clades.
- **Partition**: Channels 0–4 = awareness; channels 5–9 = aptitude.
- **Theorem suite**: 10 theorems (T-AW-1 through T-AW-10), 55/55 subtests, 0 failures.
- **Test harness**: 67 pytest cases (`test_251_awareness_cognition_closures.py`).
- **No mid-episode edits**: channel definitions, entity values, and partition boundaries are frozen for the duration of this analysis.

4. Integrity Ledger (Reconcile)

The ledger debits drift and roughness and credits return:

$$\Delta\kappa = R \cdot \tau_R - (D_\omega + D_C), \tag{A1}$$

where $D_\omega = \Gamma(\omega) = \omega^p/(1 - \omega + \varepsilon)$ is the drift cost, $D_C = \alpha \cdot C$ is the curvature cost, and $R \cdot \tau_R$ is the return credit.

For the awareness–cognition domain:

- All 34 entities have $\tau_R = \infty_{\text{rec}}$ (no return to Stable), so $R \cdot \tau_R = 0$: no credit.
- The budget residual is therefore $\Delta\kappa = -(D_\omega + D_C) < 0$ for all entities.
- The seam residual $|s| = |\Delta\kappa_{\text{budget}} - \Delta\kappa_{\text{ledger}}|$ is verified $\leq \text{tol}_{\text{seam}} = 0.005$ for all 34 kernel evaluations.

Tier-1 identity residuals across the full catalog:

Identity	Max residual
$F + \omega - 1$	$< 10^{-16}$
$\text{IC} - \exp(\kappa)$	$< 10^{-16}$
$F - \text{IC} \geq 0$ (integrity bound)	Holds for 34/34

5. Stance (Read)

The stance is derived from gates, never asserted:

Regime	Count	Condition
Stable	0/34	All four gates satisfied
Watch	0/34	$0.038 \leq \omega < 0.30$
Collapse	34/34	$\omega \geq 0.30$
Critical overlay	15/34	IC < 0.30

Verdict: The awareness–cognition closure is CONFORMANT. All Tier-1 identities hold to machine precision. All regime classifications follow from the frozen gates. The seam residuals reconcile. No Tier-1 symbol is redefined; no diagnostic is used as a gate; no back-edge exists from Tier-2 to Tier-0 or Tier-1.

Appendix B: Seam Lineage

The awareness–cognition closure is the fifth Tier-2 domain validated against the GCD kernel. Its seam lineage — the chain of validated closures that precede it and the structural results it inherits — is recorded below.

1. Lineage Chain

1. **Seam 0: Kernel Identity Proof** (`tier1_proof.py`).
   Anchor: 10,162 tests, 0 failures.
   $F + \omega = 1$ (exact, 10^{-16}); IC $\leq F$ (100%, all domains); IC $= e^{\kappa}$ (exact, 10^{-16}).
   This seam establishes that the kernel function is internally consistent before any domain closure is applied.

2. **Seam 1: Standard Model** (`closures/standard_model/`).
   Anchor: 10/10 SM theorems, 74/74 subtests.
   31 particles → 8-channel trace. Key result: confinement appears as IC cliff at the quark→hadron boundary (IC drops 98.1%). Heterogeneity gap Δ is driven by the color channel at ε.
   Seam crossing: T-AW-5 inherits the confinement signature from this closure.

3. **Seam 2: Atomic Physics** (`closures/atomic_physics/`).
   Anchor: 118 elements, 12-channel cross-scale kernel.
   Tier-1 identities verified for all 118 elements.
   Seam crossing: establishes that the kernel operates at the atomic scale with the same algebra.

4. **Seam 3: Kinematics** (`closures/kinematics/`).
   Anchor: Trajectory phase-space analysis.
   Seam crossing: T-AW-9 uses the evolution kernel (which shares organisms with the kinematics domain) to verify cross-domain F-ranking preservation.

5. **Seam 4: Evolution** (`closures/evolution/`).
   Anchor: 40 organisms, 10-channel brain kernel, 20 species comparative neuroscience.
   Seam crossing: T-AW-9 directly compares four shared organisms between the evolution and awareness kernels. F-ranking preserved ($\rho = 0.40$) despite completely different channel definitions.

6. **Seam 5: Awareness–Cognition** (`closures/awareness_cognition/`) — *this work*.
   Anchor: 34 organisms, 10-channel 5+5 partition.
   10/10 theorems, 55/55 subtests, 67 pytest cases.
   New structural results: awareness-aptitude inversion (T-AW-1), universal instability (T-AW-2), binding gate transition (T-AW-8). Inherits confinement analogy from Seam 1 and cross-domain bridge from Seam 4.

2. κ-Continuity Across the Lineage

Each seam crossing preserves κ-continuity: the log-integrity residual at each boundary satisfies $|s_k| \leq \mathrm{tol}_{\mathrm{seam}} = 0.005$. The seam chain accumulates additively (Lemma 20 of Ref. [3]):

$$\Delta\kappa_{\mathrm{total}} = \sum_{k=0}^{K} \Delta\kappa_k, \qquad \sum_{k=0}^{K} |s_k| \leq (K+1) \cdot \mathrm{tol}_{\mathrm{seam}}. \tag{B1}$$

For $K = 5$ seams (Seam 0 through Seam 5), the maximum cumulative residual is bounded by $6 \times 0.005 = 0.030$. The seam composition is associative (Lemma 45) and possesses an identity element (Lemma 46: the zero-residual seam where $\Delta\kappa = 0$).

3. Structural Inheritance

The following structural results propagate *forward* through the lineage (each entry lists the originating seam and the theorem in this work that inherits it):

Inherited result	Source	Used by
Confinement as IC cliff	Seam 1 (T3)	T-AW-5
$\mathrm{dIC}/\mathrm{d}c_k = \mathrm{IC}w_k/c_k$	Seam 0 (Lemma)	T-AW-4
$F + \omega = 1$	Seam 0 (Tier-1)	T-AW-10
IC $\leq F$	Seam 0 (Tier-1)	T-AW-10
Cross-domain ranking	Seam 4 (Evo)	T-AW-9

No structural result flows *backward*: the awareness closure does not modify any prior seam's thresholds, constants, or identity proofs. The one-way dependency Tier-1 → Tier-0 → Tier-2 is preserved.

4. Weld Certificate

Closure: `awareness_cognition` (Tier-2)
Contract: UMA.INTSTACK.v1 (`frozen_contract.py`)
Kernel: $K\colon [0,1]^{10} \times \Delta^{10} \to (F, \omega, S, C, \kappa, \mathrm{IC})$
Channels: 10 (5 awareness + 5 aptitude), $w_i = 1/10$
Entities: 34, 12 clades
Theorems: 10/10 proven, 55/55 subtests
Tests: 67 pytest cases
Identities: $F + \omega = 1$ ($< 10^{-16}$), $\mathrm{IC} \leq F$ (34/34), $\mathrm{IC} = e^{\kappa}$ ($< 10^{-16}$)
Seam residuals: $|s| \leq 0.005$ (34/34)
Regime: 34/34 Collapse, 15/34 Critical overlay, 0/34 Stable
Verdict: CONFORMANT
Lineage: Seam 0 (Tier-1 proof) → Seam 1 (SM) → Seam 2 (Atomic) → Seam 3 (Kin) → Seam 4 (Evo) → Seam 5 (this work)
Back-edges: None
Symbol capture: None

Historia numquam rescribitur; sutura tantum additur.

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